

GHOST PREDATOR — Instant Ladder Resolution

Provisional close-verdicts so the Paroli ladder steps in 2 seconds, not 7 minutes · build guide for the team · June 11, 2026

The Problem: the ladder lags the chain

Polymarket settles each Up/Down window ON-CHAIN ~5–7 minutes after close. If your Paroli ladder only counts officially-resolved trades, it is permanently 1–2 windows behind on 5m markets: wins don't climb the stake in time, losses don't reset it in time, and the bot "gets stuck shooting the same number." Real examples from our book: a trade bet base-\$10 right after TWO wins because those wins hadn't settled yet (they confirmed 3 minutes later); a \$22.50 top rung carried through a 2-hour trading pause because the streak couldn't update. The Polymarket SITE shows results instantly because win/loss is knowable the second the window closes — final price vs price-to-beat. Only the payout needs the chain. So judge the outcome yourself, immediately, from your own price feed — and use that verdict for STAKE SIZING ONLY.

Design Rule #1 (do not skip this)

The provisional verdict drives the ladder step and NOTHING else. Official on-chain resolution remains the only authority for: recorded win/loss status, PnL, balance, and every guardrail (loss caps, streak halt, balance floor). If your tape is missing at the close, or the close exactly ties the open — write NO verdict and let the chain decide. Worst case of a wrong provisional call (razor photo-finish where your feed and the Chainlink oracle disagree): the ladder bets one wrong rung for ~5 minutes, then the chain overwrites and it self-corrects. Bounded, rare, acceptable. Our live mismatch rate so far: 0 of 7.

How It Works (3 pieces)

1. A per-second price tape. You already have a live Binance trade feed; keep the last ~15 minutes of 1-second closes per coin in memory (a deque). The LAST trade price of each second is the close tick.

```
px_sec = {} # COIN -> deque[(epoch_sec, last_trade_px)], maxlen=900
# inside your Binance trade handler, after updating the live price:
sec = int(now())
dq = px_sec.setdefault(coin, deque(maxlen=900))
if dq and dq[-1][0] == sec: dq[-1] = (sec, px) # keep LAST trade of the second
else: dq.append((sec, px))
```

2. A provisional-verdict loop (1-second async tick). For every open position whose window closed ≥ 2 s ago and has no verdict yet: look up the tape price AT the close second, compare to the stored window-open price, write 'won'/'lost' to a separate prov column (add it via migration; never touch status).

```
async def prov_loop():
    while True:
        for pid, coin, side, open_px, close_ts in open_positions_without_prov(close_ts <= now()-2):
            px = last_tape_price_at_or_before(coin, close_ts) # walk px_sec backwards
            if px is None or px == open_px: continue # no tape / exact tie -> chain decides
            up_won = px > open_px
            prov = 'won' if (side in ('Up', 'Yes')) == up_won else 'lost'
            db_update("UPDATE positions SET prov=? WHERE id=?", prov, pid)
            log(f"[prov] #{pid} {coin} {side} -> {prov} — ladder steps NOW, chain confirms in ~5min")
            await asyncio.sleep(1.0)
```

3. The ladder counts effective status. In your paroli/stake function, treat each recent trade as: official status if resolved, else the provisional verdict, else skip (truly in-flight).

```
rows = db("SELECT status, prov FROM positions WHERE status IN ('won', 'lost', 'open')
```

```

        AND mode='LIVE' ORDER BY id DESC LIMIT ?", window)
consec = 0
for st, pv in rows:
    eff = st if st in ('won', 'lost') else (pv or 'open')
    if eff == 'won': consec += 1
    elif eff == 'open': continue          # window not closed yet / no tape: never counts as a win
    else: break                          # loss (official OR provisional) -> reset to base

```

Verified Live

Event	Before	After
Ladder reacts to a win/loss	~5–7 min (on-chain settle)	~2 seconds after window close
First live verdict	—	trade closed 04:25:00, [prov] logged 04:25:02
Prov vs chain agreement	—	7 / 7 so far (audit query below)
Wrong-rung exposure on a bad call	—	one mis-sized bet for ~5 min, self-corrects

Audit your own mismatch rate weekly: `SELECT COUNT(*) FROM positions WHERE prov IS NOT NULL AND status IN('won', 'lost') AND prov<>status`; If it creeps above ~2–3%, your feed and the resolution oracle are diverging — investigate before trusting provisional steps.

Gotchas We Hit

- **Resolution source ≠ your feed.** Polymarket resolves on Chainlink data streams (USD); a Binance USDT feed can sit a few dollars away. That offset mostly cancels because BOTH your open and close come from the same feed — but on dead-flat windows the disagreement can flip a razor verdict. Hence: ties write nothing, and guardrails never trust prov.
- **Restart gaps:** after a bot restart the tape is empty; windows that closed during the gap simply get no provisional verdict and wait for the chain. No special handling needed.
- **Don't count in-flight as wins.** Eff-status 'open' must SKIP, not count — counting opens as wins was an earlier real bug that laddered stakes up during losing streaks.
- Pairs well with duration-weighted Paroli (cap top rungs on 5m, full ladder on 15m) — the instant verdicts make the cycle actually cycle at 5m speed.